

# Sreehari Suresh Babu

## Education

2022–Present **PhD in Mathematics (in progress)**, *University of Kansas*

2016–2021 **BS–MS Dual Degree**, *IISER Thiruvananthapuram*

Major: Mathematics; Minor: Physics;

## Publications/Preprints

2. **Linear Resolutions and Linear Quotients in Three Variables (with Hoai Dao)**, *Combinatorial Commutative Algebra*, 2026
1. **Componentwise Linear Ideals From Sums (with Hailong Dao)**

## Service

- Co-organizer, University of Kansas Student Algebra Seminar, Fall 2025, Spring 2026, Fall 2026
- Activity organizer for Young Sheldon, Summer 2026 (Organized a K-12 science outreach event in Olathe, KS promoting curiosity and hands-on exploration)
- Leader, University of Kansas Algebra Qualifying Exam Study Group, Summer 2025

## Honors and Awards

- Charles J. and Mary Pat Himmelberg Graduate Student Award, 2026, KU
- U.G. Mitchell Graduate Scholarship in Mathematics, Summer 2023, 2024, 2025, KU
- Junior Research Fellow (JRF), IISER Pune
- School Gold Medal 2021, School of Mathematics, IISER Thiruvananthapuram
- INSPIRE Fellowship from Department of Science and Technology, India

## Invited Talks

**Socles of Powers**, *AMS 2026 Fall Western Sectional Meeting*, Arizona State University, November 2026

**Componentwise Linear Ideals From Sums**, *Ideal Conversations: Student Commutative Algebra Seminar*, Purdue University, April 2026

**Socles of Powers**, *Purdue Commutative Algebra Seminar*, Purdue University, April 2026

**Componentwise Linear Ideals From Sums**, *Number Theory and Algebra Seminar*, Arizona State University, April 2026

**Componentwise Linear Ideals From Sums**, *Combinatorial and Commutative Algebra*, Oklahoma State University, October 2025

**Componentwise Linear Ideals From Sums**, *Problem-solving seminar in commutative algebra (Virtual)*, Stockholm University, April 2025

**On The Optimal Polarization of Monomial Ideals**, *AMS 2025 Spring Central Sectional Meeting*, University of Kansas, March 2025

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## Other Talks

**The Conductor of a Ring**, *Student Algebra Seminar*, University of Kansas, September 2026

**The Projective Dimension of Three-generated Ideals**, *Student Algebra Seminar*, University of Kansas, March 2026

**An Introduction to Stanley-Reisner Rings**, *Student Algebra Seminar*, University of Kansas, September 2025

**Componentwise Linear Ideals From Sums**, *Commutative Algebra and Algebraic Geometry Seminar*, Virtual, September 2025

**Groebner bases**, *GSO Seminar*, University of Kansas, February 2025

**DG-Algebras and Free Resolutions**, *Student Algebra Seminar*, University of Kansas, November 2024

**Local Cohomology and Global Cohomology**, *Student Algebra Seminar*, University of Kansas, February 2024

**Socle Degrees of Finite Sets of Points in Projective Space**, *Algebra Day at KU*, University of Kansas, November 2023

**Catelnuovo-Mumford Regularity of Subspace Arrangements**, *Student Algebra Seminar*, University of Kansas, October 2023

**p-adic Integers**, *GSO Seminar*, University of Kansas, September 2023

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## Conferences/Workshops Attended

October 2026 **Accelerating Mathematical Discovery: A Crash Course in AI and Computation**, Santa Cruz, CA

October 2026 **KUMUNU 2026**, *University of Kansas*, Lawrence, KS

August 2026 **URiCA 2026**, *University of Nebraska–Lincoln*, Lincoln, NE

March 2026 **CA+**, *Iowa State University*, Ames, IA

December 2025 **Arizona-New Mexico Symposium on Commutative Algebra and its Interactions: Geometric Combinatorics**, *New Mexico State University*, Las Cruces, NM

October 2025 **KUMUNU 2025**, *University of Missouri*, Columbia, MO

July 2025 **Macaulay2 Workshop: Madison 2025**, *University of Wisconsin - Madison*, Madison, WI

May 2025 **URiCA 2025**, *University of Nebraska–Lincoln*, Lincoln, NE

April 2025 **RayFest**, *University of Nebraska–Lincoln*, Lincoln, NE

September 2024 **KUMUNU 2024**, *University of Missouri*, Columbia, MO

August 2024 **Multigraded and differential graded methods in commutative algebra (SLMath Summer Graduate School)**, *St. Mary's College*, Moraga, CA

June 2024 **NCMW - Recent Trends in Commutative Algebra (2024)**, *IIT Bombay*, Powai

September 2023 **KUMUNU 2023**, *University of Missouri*, Columbia, MO

May 2023 **URiCA 2023**, *University of Nebraska–Lincoln*, Lincoln, NE

April 2023 **Kansas Mathematics Graduate Student Conference (KMGSC)**, *Kansas State University*, Manhattan, KS

October 2022 **KUMUNU 2022**, *University of Nebraska–Lincoln*, Lincoln, NE

August 2021 **Elliptic Curves and The Special Values of L-functions (ONLINE)**, *ICTS*, Bengaluru

Summer 2019 **Summer Fellow**, *IIT Madras*, Chennai  
Title: *Vanishing of Matrix Coefficients at infinity*; Advisor: Dr. Aprameyan P  
December 2016 **National Science Camp 2016 (VIJYOSHI)**, *IISc.*, Bangalore

## Teaching

Fall 2026 **MATH 115: Calculus I**, *University of Kansas*  
Instructor  
Summer 2026 **MATH 115: Calculus I**, *University of Kansas*  
Instructor  
Spring 2026 **MATH 127: Calculus III**, *University of Kansas*  
Lab Recitation  
Fall 2025 **MATH 790: Linear Algebra II**, *University of Kansas*  
Grader  
Fall 2025 **MATH 145: Calculus I, Honors**, *University of Kansas*  
Lab Recitation  
Summer 2025 **Algebra Qual Study Group**, *University of Kansas*  
Leader  
Spring 2025 **MATH 126: Calculus II**, *University of Kansas*  
Lab Recitation  
Fall 2024 **MATH 126: Calculus II**, *University of Kansas*  
Lab Recitation  
Spring 2024 **MATH 115: Calculus I**, *University of Kansas*  
Instructor  
Fall 2023 **MATH 115: Calculus I**, *University of Kansas*  
Instructor  
Spring 2023 **MATH 101: College Algebra**, *University of Kansas*  
Instructor  
Fall 2022 **MATH 002: Intermediate Mathematics**, *University of Kansas*  
Instructor

## Others

Languages Malayalam (Native), English (Fluent), Hindi (Fluent), French (Reading proficiency)  
Skills Python, Macaulay2,  $\text{\LaTeX}$